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Interfacial nanostructure and friction of a polymeric ionic liquid-ionic liquid mixture as a function of potential at Au(111) electrode interface

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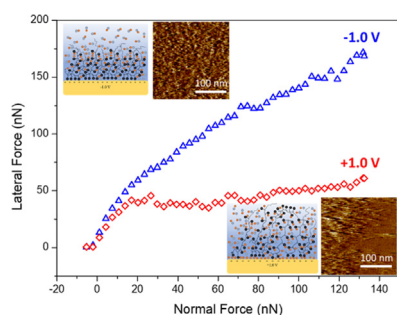
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GRAPHICAL ABSTRACT



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ABSTRACT

Hypothesis: The polymeric cations of polymeric ionic liquids (PILs) can adsorb from the bulk of a conventional ionic liquid (IL) to the Au(111) electrode interface and form a boundary layer. The interfacial properties of the PIL boundary layer may be tuned by potential.

Experiments: Atomic force microscopy has been used to investigate the changes of surface morphology, normal and lateral forces of a 5 wt% PIL/IL mixture as a function of potential.

Findings: Polymeric cations adsorb strongly to Au(111) and form a polymeric cation-enriched boundary layer at -1.0 V. This boundary layer binds less strongly to the surface at open circuit potential (OCP) and weakly at $+1.0$ V. The polymeric cation chains are compressed at -1.0 V and OCP owing to electrical attractions with the electrode surface, but fully stretched at $+1.0$ V due to electrical repulsions. The lateral forces of the 5 wt% PIL/IL mixture at -1.0 V and OCP are higher than at $+1.0$ V as the polymeric cation-enriched boundary layer is rougher and has stronger interactions with the AFM probe; at $+1.0$ V, the lateral force is low and comparable to pure conventional IL due to displacement of polymeric cations with conventional anions in the boundary layer.

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Abbreviations: PILs, polymeric ionic liquids; IL, ionic liquid; OCP, open circuit potential; AFM, atomic force microscopy; Poly(3MAPIm) TFSI, poly(3-methyl-1-aminopropylimidazolylacrylamide) bis(trifluoro-methylsulfonyl)imide; BMIM TFSI, 1-butyl-3-methylimidazolium bis-(trifluoromethylsulfonyl)imide; THF, tetrahydrofuran; DMT, Derjaguin–Müller–Toporov model; JKR, Johnson–Kendall–Roberts model; F_A , adhesive forces; F_N , normal force; $Ex(F_N)$, the apparent separation at a given normal force; L_{contour} , contour length; L_K , Kuhn length; F_{LT} , total lateral force; S_c , shear strength; a , surface contact radius; μ , friction coefficient.

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1. Introduction

Interfacial properties, including adhesion and friction, are critical for processes involving surface contact and relative motion. Lubricants are applied to surfaces and act as barriers to reduce friction during sliding.[1] About a quarter of the global energy consumption originates from the energy losses in friction and wear

[2], thus there is a constant need to develop new and more efficient lubricants. Over the last two decades, the study of friction and lubrication at the nanoscale has increased dramatically, driven in part by the advent of micro and nano-electromechanical systems [3–6]. At the nanoscale, friction and energy dissipation are reduced due to the formation of a strongly adsorbed lubricant boundary layer on the sliding surfaces [6–8], thus it is crucial to understand adsorption mechanisms and boundary layer properties for the design of high-performing lubricants.

Ionic liquids (ILs) are pure salts with melting points below 100 °C. [9–11] ILs have been widely investigated as electrolytes in batteries and supercapacitors due to their superb electrical conductivity and stability. [12–14] They have also been studied as excellent lubricant candidates owing to their high temperature stability and low vapour pressure. [15–19] One distinguishing feature of IL lubricants from other high-performing lubricants is that the friction of ILs can be actively controlled via tuning the interfacial nanostructure of the IL, especially in the boundary layer, using an external electric potential, even when it is diluted in oil. [20–23] However, the magnitude of the differences in friction with potential using conventional ILs is usually small, [23–26] which limits their applications in industry.

Polymeric ionic liquids (PILs) are a new class of ILs which comprise a polymer backbone containing tethered cations (or anions) and associated counterions. [27–28] They combine the attractive properties of ILs with the sequence complexity and the mechanical characteristics of macromolecules. As flexible, gel-like materials, PILs are excellent candidates for flexible thin film electrolytes due to their good conductivity originated from the mobility of counterions and excellent mechanical characteristics due to extended polymer backbone (with associated cation pendant groups). [29] PILs also have a variety of properties that allow them to be applied as potential lubricants under extreme conditions. [30] For example, PILs are electrically conductive and are thus suitable for electrical applications. PILs can prevent electrical arcing, which causes decomposition of sliding contacts. PILs are better able to dissipate heat, as they have even higher thermal conductivity than that of high-performing synthetic lubricants such as perfluoropolyethers. Additionally, as the PIL ions interact strongly with oppositely charged surfaces, they could resist ‘squeeze out’ similar as conventional ILs when surfaces are compressed. [31] However, to date the interfacial properties of PILs at electrode interfaces, including interfacial nanostructure and nanofriction have not been investigated despite their importance for electrochemical and lubrication applications.

The interfacial nanostructures of ILs and related materials have been investigated using atomic force microscopy (AFM). [32–37] In a typical AFM normal force vs separation measurement, step-wise profiles are detected when the AFM probe approaches the surface due to push through of well-packed ion/ion pair layers. [38] The ‘push-through force’ for an IL layer is reflected by the maximum of each step; higher push-through forces indicate higher ordering of ions and stronger cohesive interactions within the layer. The separations between two adjacent steps are generally consistent with the dimensions of single ion or ion pairs, which provide information on the composition of each layer. AFM is also an excellent technique to investigate friction at the nanoscale, as AFM can measure both the normal force and lateral (frictional) force at the same time via tracking the normal and lateral deflections of the AFM probe. [23,25] Lateral force generally increases linearly with normal force when the adhesion between the two contacting surfaces is weak. The proportionality between the lateral force and the normal force in the linear regime is given by the friction coefficient (μ), which is a useful parameter in comparing the lubrication properties between systems. [39]

When dissolved in ILs, polymers have been found to form robust adsorbed polymer layers at solid surfaces and reduce surface contact and friction while maintaining the useful properties of ILs as lubricants. [40] Therefore, in order to combine the useful properties both PILs and conventional ILs in this project, we dissolve 5 wt% of one PIL, poly(3-methyl-1-aminopropylimidazolylacrylamide) bis(trifluoro-methylsulfonyl)imide (Poly(3MAPIm) TFSI), in one conventional IL, 1-butyl-3-methylimidazolium bis-(trifluoromethylsulfonyl)imide (BMIM TFSI), cf. Fig. 1, and use AFM to investigate the morphology, normal nanostructure and friction of the adsorbed PIL as a function of potential at an Au(111) electrode surface. The outcomes help to elucidate the interfacial adsorption mechanisms of PIL electrolytes at the Au(111) electrode towards bridging the gap between conventional liquid electrolytes and flexible solid state electrochemical devices.

2. Materials and methods

2.1. Materials

BMIM TFSI (99.5%) was purchased from Ionic Liquids Technologies (Germany). Tetrahydrofuran (THF) (99.9%) was purchased from Sigma-Aldrich. The PIL Poly(3MAPIm) TFSI with a degree of polymerisation of 248 and polydispersity index of 1.13 was synthesised at University of California, Santa Barbara (UCSB), as described

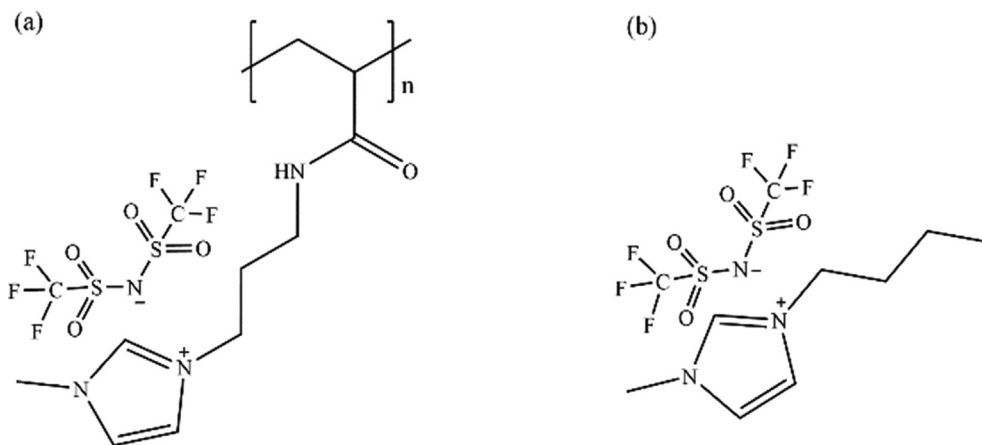


Fig. 1. Chemical structures of (a) the polymeric ionic liquid Poly(3MAPIm) TFSI and (b) the conventional ionic liquid BMIM TFSI, respectively.

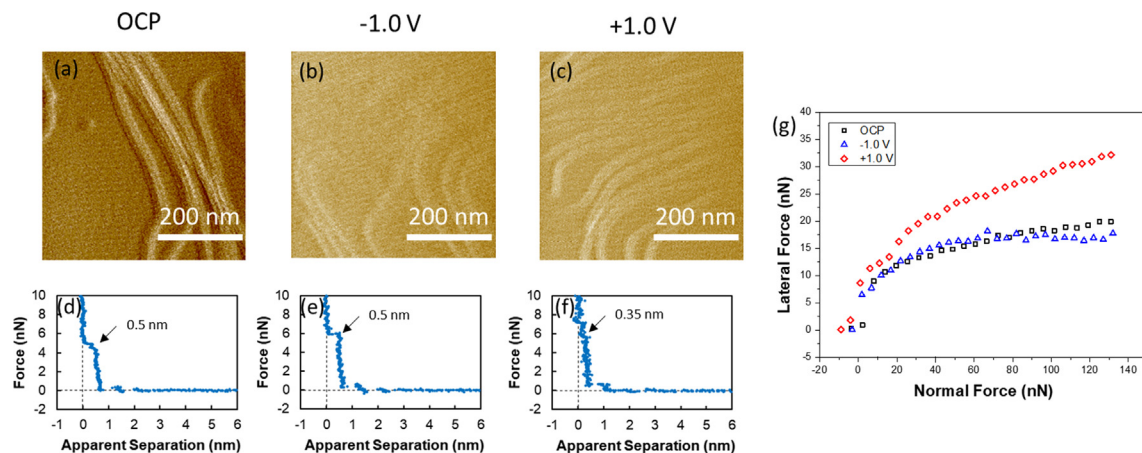


Fig. 2. The surface morphology, interfacial nanostructure and friction of pure BMIM TFSI on Au(111) at different potentials. (a)–(c): AFM contact mode deflection images of BMIM TFSI on Au(111) at OCP, -1.0 V and $+1.0$ V, respectively. The height scales from topographic images are between ± 1 nm. The normal force applied on the AFM probe to achieve these images is 32 nN. (d)–(f): normal force vs apparent separation data for the approach of an AFM probe to Au(111) immersed in pure BMIM TFSI at OCP, -1.0 V and $+1.0$ V, respectively. (g): lateral force vs. normal load of BMIM TFSI on Au(111) at OCP, -1.0 V and $+1.0$ V, respectively.

in detail elsewhere.[29] Atomically smooth Au(111) surfaces (a gold film of 300 nm thickness on mica) were purchased from Georg Albert PVD – Beschichtungen.

2.2. Preparation of PIL-IL mixtures

Since pure PIL is paste-like and difficult to mix evenly with BMIM TFSI directly, 5 wt% PIL/BMIM TFSI solution was made as follows. 0.2 g PIL was firstly dissolved in 2.0 g tetrahydrofuran (THF), then mixed with 3.8 g BMIM TFSI. After mixing, the sample was covered with filter paper and left overnight to evaporate most of the THF. The remaining THF was then removed under vacuum at 100 °C for 24 h.

2.3. AFM measurements

AFM imaging, normal force curve and friction measurements were performed using a Veeco Nanoscope IV AFM in the contact mode on Au(111) as a function of potential. Sharp silicon AFM tips (NSC36, Mikromasch) with spring constants of 0.8 ± 0.3 N/m and ~ 8 nm probe radius and the Au(111) surface were cleaned with MilliQ water and ethanol, followed by UV-Ozone for 10 min before use. The AFM electrochemical cell (Bruker) was washed with MilliQ water followed by ethanol, and dried in nitrogen flow before experiment. Au(111) surface was used as the working electrode; 0.25 mm Pt wires were used both as the counter and “quasi” reference electrodes.

AFM experiments were conducted until the system reached quasi-equilibrium so as to minimize thermal drifts. The AFM cantilever deflections were calibrated on Au(111) in air and in pure BMIM TFSI before measurements in 5 wt% PIL/BMIM TFSI. The topography and normal force curves of Au(111) in air and in pure BMIM TFSI are shown in Figure S1, and Fig. 2, respectively. AFM images were collected at a series of deflection setpoints with scan lines of 256×256 and scan rates of 6 Hz. At this scan rate, the scan time is significantly reduced, whereas the major features remain sharp and clear in the images. Normal force curves were measured by moving the Au(111) surface towards the AFM probe and detecting the cantilever deflection as a function of separation. The ramp sizes were 50–150 nm and ramp rate was 0.2 Hz. Standard methods were used to convert deflection vs separation data to normal force vs separation curves.[41] Lateral forces were measured by performing AFM scans with a scan angle of 90° (with respect to the long axis of the cantilever) and with the slow scan axis dis-

abled.[41] The scan size was 500 nm; scan rate was 6 Hz. The lateral deflection signal (i.e. cantilever twist) was converted to lateral force using a customized function produced in MATLAB which takes into account the torsional spring constant and the geometrical dimensions of the cantilever.[42] Every experiment was repeated at least three times.

3. Results and discussion

3.1. Interfacial nanostructure of friction of pure BMIM TFSI

Fig. 2 (a)–(c) shows the morphologies of Au(111) in pure BMIM TFSI at different potentials. As the potential changes from OCP to -1.0 V and then increased to $+1.0$ V, the morphology of Au(111) does not alter significantly from that in air (Figure S1) and shows clear step-edges created from manufacturing process. This suggests no obvious surface restructuring upon immersing Au(111) in pure BMIM TFSI. The AFM normal force–apparent separation profiles for pure BMIM TFSI (Fig. 2 (d)–(f)) consist of a series of pronounced, discrete steps, which are produced by the AFM probe pushing up against interfacial layers, similar to those seen previously.[43] For pure BMIM TFSI, the innermost push-through forces at both ± 1.0 V are higher than at OCP, suggesting that BMIM TFSI develops stronger interfacial structure when the surface is biased than when it is neutral. The layer thicknesses of the innermost push-through layers at -1.0 V and OCP are ~ 0.5 nm, consistent with the dimension of a TFSI anion. This means there exists at least one cation-enriched boundary layer that could not be pushed through and thus detected by the AFM probe up to high force, also as seen before.[43] At $+1.0$ V, the thickness of the final push-through layer (~ 0.35 nm) is comparable with the dimension of a BMIM cation, thus suggests that an anion boundary layer binds strongly to the positively charged Au(111) surface and could not be penetrated. At all three potentials, the lateral (frictional) force increases steeply when the normal force is below 10 nN and then more gradually when the normal force increases further, as shown in Fig. 2(g). This suggests the AFM probe slides across the relatively rough and less ordered IL multilayers in the low normal force regime followed by the well-defined boundary layer in the high normal force regime. The lateral forces at -1.0 V and OCP are lower than those at $+1.0$ V. This indicates that the final, impenetrable cation-enriched boundary layer is more robust than the anion-enriched boundary layer, thus lubricating the Au(111) surface better, in line with previous studies.[23–24]

3.2. Surface morphologies of 5 wt% PIL/BMIM TFSI on Au(111) at different potentials

The surface morphologies of 5 wt% PIL/BMIM TFSI on Au(111) at different potentials are detected using contact mode AFM, as shown in Fig. 3. The deflection images are selected as they present surface features more clearly; the topographic images with height scales are shown in Figure S2. In contact mode, the normal forces applied on the AFM probe during scanning can be calculated from the deflection setpoint, as described in detail in the SI. For the images in the left column, a low normal force of 10 nN is applied. The AFM probe scans harder across the surface at a normal force of 32 nN for the images in the middle column and hardest at a normal force of 64 nN for the images in the right column.

For 5 wt% PIL/BMIM TFSI at OCP (Fig. 3(a)), when the AFM probe contacts the surface gently, there likely exists a weakly adsorbed layer which masks the original step-edges observed on bare Au(111) (Figure S1). The adsorbed layer appears to be dragged by the AFM probe as it scans back and forth across the surface. As the AFM probe scans the surface with greater applied force, the adsorbed layer is scraped away, and the underlying step-edges again become visible (Fig. 3 (b) and (c)). The displaced adsorbed layer reappears after 30 min (Figure S3). These results strongly suggest that the PIL adsorbs from the bulk liquid to the Au(111) surface and forms a polymeric cation-enriched boundary layer. As the Au(111) was cleaned using UV-Ozone before the AFM experiment, the surface is prone to be hydrophilic and slightly negatively charged in the BMIM TFSI solvent, the polymeric cations

tend to adsorb with the charged pendent groups towards the surface and the uncharged polymer backbones towards the bulk, as illustrated in Fig. 4 (a). However, at OCP, polymeric cations are only weakly adsorbed to Au(111) likely due to weak electrostatic interactions with the Au(111) surface and competitive adsorption of BMIM cations from the BMIM TFSI solvent. Thus, the adsorbed polymeric cation-enriched boundary layer can be removed when the AFM probe scans harder across the surface. At -1.0 V, stronger electrostatic interactions between the polymeric cations and the negatively charged Au(111) surface lead to the polymeric cation-enriched layer being adsorbed to the surface more strongly so that it could no longer be removed by increasing the normal force applied on the AFM probe during scanning, as imaged in Fig. 3 (d)–(f). Compared to OCP, this adsorbed polymeric cation layer looks more compact, and is not dragged during the movement of the AFM probe, indicating that the polymeric cations pack more densely at -1.0 V than at OCP, as sketched in Fig. 4 (b). In contrast, at $+1.0$ V, the adsorbed layer looks “fluffy” and could be brushed aside even when the AFM probe contacts the surface gently, cf. Fig. 3 (g)–(h). This is similar to previous reports of removing polymer coatings from solid substrates by scrubbing.[44] The thickness of the adsorbed layer is ~ 3.3 nm, cf. Figure S4; the value is consistent with the radius of gyration of the same PIL in the BMIM TFSI obtained using small angle neutron scattering, which is the topic of a forthcoming article. This adsorbed polymeric cation layer could be removed easily because the polymeric cations are repelled from the positively charged surface, leading to weakest adsorption, as illustrated in Fig. 4 (c).

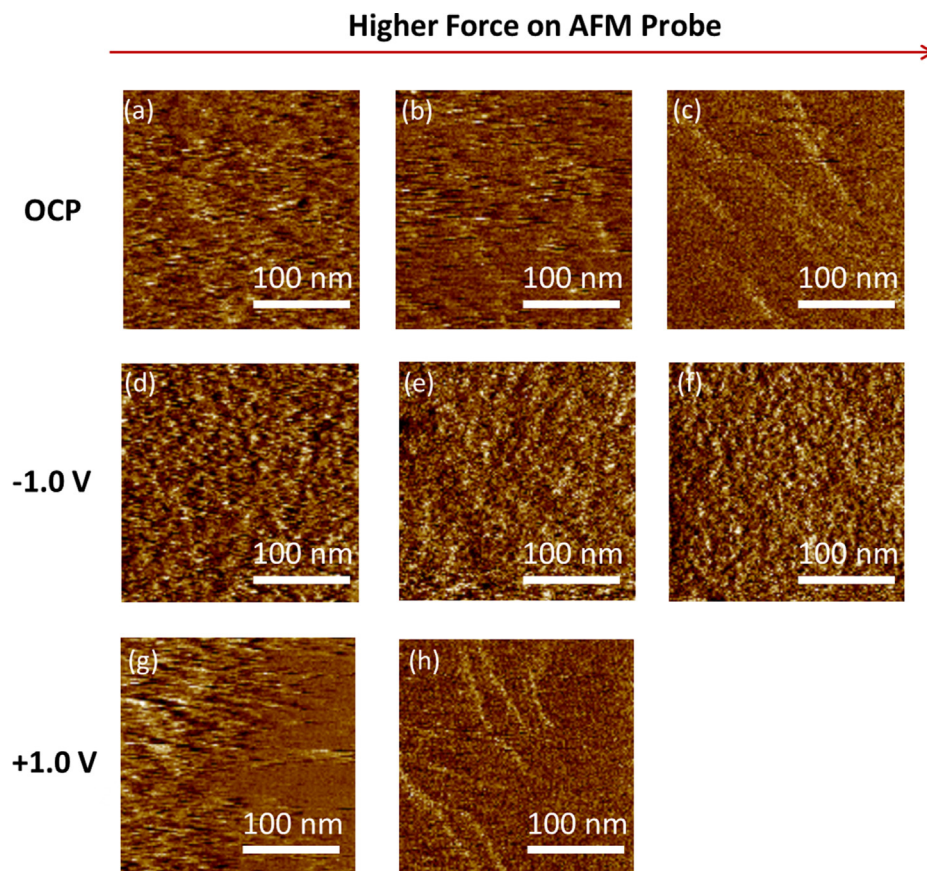


Fig. 3. AFM contact mode deflection images of 5 wt% PIL/BMIM TFSI on Au(111) at different potentials and normal forces. (a)–(c): normal force increases from 10 nN to 32 nN and 64 nN when Au(111) is at OCP. (d)–(f): normal force increases from 10 nN to 32 nN and 64 nN when Au(111) is -1.0 V. (g)–(h): normal force increases from 10 nN to 32 nN when Au(111) is $+1.0$ V.

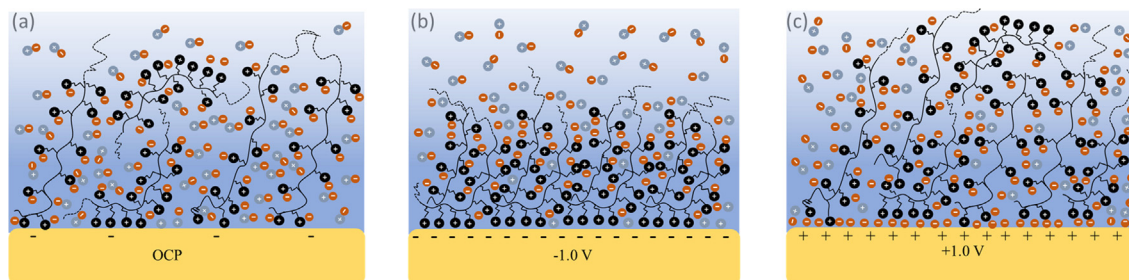


Fig. 4. Schematic of the adsorbed boundary layer of 5 wt% PIL/BMIM TFSI on Au(111) at different potentials. (a) OCP, (b) -1.0 V, and (c) $+1.0$ V, respectively. The positively charged pendent group of the PIL is dark blue, the BMIM cation is light blue and the TFSI anion is orange. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.3. Normal force curves of 5 wt% PIL/BMIM TFSI on Au(111) at different potentials

Representative normal force–separation profiles for the approach of an AFM probe to an Au(111) surface immersed in 5 wt% PIL/BMIM TFSI are shown in Fig. 5 (a)–(c). At both OCP and -1.0 V, unlike those detected in pure BMIM TFSI, no step-wise forces are detected. Instead, the normal force increases smoothly and monotonically from zero at a separation of ~ 3 nm to 30 nN at the zero apparent separation. These results are consistent with the obtained AFM images (Fig. 3 (a)–(f)) and indicate that for 5 wt% PIL/BMIM TFSI, the polymeric cations adsorb to both the neutral and negatively charged Au(111) surface more strongly than BMIM cations, forming a polymeric cation enriched boundary layer, which can be compressed by the AFM probe, similar to other polymers.[45–46] Rare ion/ion pair layers are detected on top of this polymeric cation-enriched boundary layer as indicated by the circles and arrows in Fig. 5, likely caused by increased surface roughness and also reduced overscreening effects of polymeric cation boundary layer compared to conventional ILs. At $+1.0$ V, two types of normal force curves are detected. The majority of the normal force curves ($\sim 80\%$) show a similar compression profile as at OCP and -1.0 V, but with obviously lower final (contact) force (~ 17 nN) (Fig. 5 (c)); about 20% of the normal force curves show

step-wise profiles which resemble push-through forces of IL solvent layers rather than a compression force of polymers (Fig. 5 (c) inset). These results are consistent with the facile removal of a polymeric cation-enriched layer during imaging by the AFM probe under low normal force, as shown in Fig. 3 (g)–(h).

As the AFM probe retracts from the Au(111) surface immersed in 5 wt% PIL/BMIM TFSI, saw-tooth structured adhesive forces (F_A) are detected at all potentials (Fig. 5 (d)–(f) and Figure S5), while no such adhesive forces are observed in pure BMIM TFSI (Figure S6). Such saw-tooth profiles may result from the extension and detachment of multiple polymer chains, or ensembles of polymer chains bridging between the polymeric cations adsorbed on Au(111) surface and the silica AFM probe, consistent with previous studies.[47–52] Similar but much weaker saw-tooth structured adhesive forces are observed for the same 5 wt% PIL/BMIM TFSI adsorbed on a silica surface when using the same silica AFM probe (Figure S7), which indicates that the polymeric cations adsorb more strongly to the Au(111) surface than the silica probe in this study. The following results for saw-tooth structured adhesive forces are based on multiple data sets, c.f. Fig. 5 and Figure S5. The maximum adhesive forces on Au(111) at OCP and -1.0 V are larger than at $+1.0$ V. This is because polymeric cations are more enriched and densely-packed in the boundary layer at OCP and -1.0 V than at $+1.0$ V, leading to greater entanglement of polymeric chains as

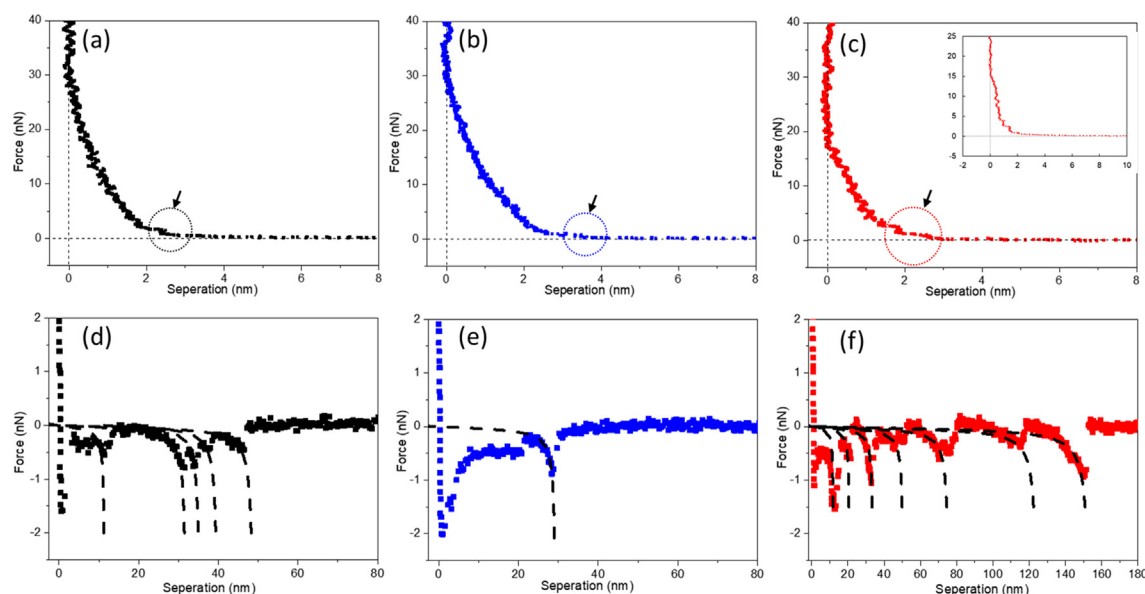


Fig. 5. Representative normal force vs apparent separation data for (a)–(c) the approach and (d)–(f) the retraction of an AFM probe from an Au(111) surface immersed in 5 wt% PIL/BMIM TFSI at different potentials. (a, d): OCP, (b, e): -1.0 V, and (c, f): $+1.0$ V, respectively. The circles and arrows in (a)–(c) indicate the detected weak ion/ion pair layers. Inset of (c): step-wise profile of PIL approach normal force curves at $+1.0$ V. Dashed lines in (d)–(f) are fits to the modified freely-jointed chain model for the retraction force data. Fit parameters are shown in Table 1.

well as more attachment points between the polymeric cations and the AFM probe.[53] At + 1.0 V, the saw-tooth structure was observed at even larger separation distances and with more bridges (7 ± 3) than OCP (5 ± 2) and -1.0 V (1 ± 1). This suggests easier extension of the polymeric cation chains at + 1.0 V, likely caused by charge repulsions between the positively charged Au (111) surface and the polymeric cations. This result is in line with the AFM images in Fig. 3 (g)–(h) which show the most fluffy-like and weakly adsorbed boundary layer at + 1.0 V.

To gain insight into the polymeric cation chain stretching, retraction force curves of 5 wt% PIL/BMIM TFSI at different potentials (Fig. 5 (d)–(f)) are fitted with a modified freely-jointed chain model in which the polymer is modelled as a chain of orientationally-independent Kuhn segments[54–55]:

$$Ex(F_N) = L_{\text{contour}} \left[\coth \left(\frac{F_N \times L_K}{k_B T} \right) - \frac{k_B T}{F_N \times L_K} \right] \quad (1)$$

where $Ex(F_N)$ is the apparent separation at a given normal force (F_N), L_{contour} is the contour length of the polymer bridge between the AFM probe and the surface, k_B is the Boltzmann constant, T is absolute temperature and L_K is the Kuhn length, which is related to the persistence length. Fits to retraction data obtained are presented as dashed lines in Fig. 5 (d)–(f), and the fitted L_{contour} and L_K values are listed in Table 1.

The contour length of the PIL, estimated from the degree of polymerisation of the polymer backbone, is ~ 63 nm. Although multiple polymeric cation chains are likely attached to the AFM probe as it retracts from the surface, the last bridge furthest away from the surface in the retraction force curve is most likely due to a single polymeric cation chain being forced to adopt an energetically less favourable extended conformation as the AFM retracts the probe from the surface.[56] The contour lengths fitted from these last bridges for both OCP and -1.0 V are clearly smaller than the estimated contour length of the PIL, suggesting that the polymeric cation chains attach strongly to the Au(111) surface, and detach from the retracting AFM probe before reaching their fully-extended contour length.

Similar features are also seen at + 1.0 V, but here retractions are also found with fitted contour lengths up to 154 nm (c.f. Table 1). This is too long to be explained by polydispersity of the PIL, instead indicating that two (or more) interacting polymeric cation chains are fully stretched as they bridge the AFM probe retracting from the positively charged surface. This is enabled by the weaker adsorption of the polymer cation on the positively charged gold and consistent with the less dense adsorbed layer.

In all cases the fitted Kuhn lengths are similar in length to a C–C bond. This result is in line with previous studies of polymers with similar backbones dissolved in ILs, [56–58] and indicates that the polymeric cation chains are extremely flexible when dissolved in BMIM TFSI.

3.4. Lateral (Frictional) force curves of 5 wt% PIL/BMIM TFSI on Au (111) at different potentials

Fig. 6 (a) shows the lateral forces versus normal forces for 5 wt% PIL/BMIM TFSI as a function of potential. At + 1.0 V, the lateral force of 5 wt% PIL/BMIM TFSI is much lower than OCP and -1.0 V and

more comparable with pure BMIM TFSI of the same potential in Fig. 2(g). A bump at a normal force of 20 nN appears as the normal force increases from 0 nN to 140 nN (loading), but disappears when the normal force decreases from 140 nN to 0 nN (unloading), as seen in Fig. 6 (b). The push-through force required to rupture the innermost layer for 5 wt% PIL/BMIM TFSI at + 1.0 V is ~ 15 nN (Fig. 5 (c) inset), which is roughly consistent with the normal force at which the bump is observed. Both results strongly support the findings from AFM imaging (Fig. 3 (g)–(h)) that the polymeric cation-enriched boundary layer is weakly adsorbed at the positively charged surface and could be easily brushed away by the AFM probe at a low normal force (~ 20 nN). The lateral force increases more slowly and is comparable with pure BMIM TFSI after the removal of the polymeric cation layer, likely because in this regime TFSI anions from either the BMIM TFSI solvent or the PIL could refill the boundary layer and lubricate the Au(111) surface, thus show comparable friction with pure BMIM TFSI. When the AFM probe scans across the same position of Au(111) for the second time, a less obvious bump is observed but shifts to a lower normal force regime (Fig. 6 (b)). This clearly shows the reproducibility of the first-time loading curve, and indicate that the adsorbed polymeric cation layer is difficult to reform once destroyed at + 1.0 V.

At -1.0 V and OCP, the lateral forces of 5 wt% PIL/BMIM TFSI are more than five times higher than those for pure BMIM TFSI (Fig. 2 (g)). This means that the adsorbed polymeric cation-enriched boundary layer does not assist in reducing friction, likely due to increase in surface roughness caused by compression of polymeric cation chains and stronger attractions between the polymeric cations and the silica AFM probe, as indicated the sawtooth retractions in Fig. 5.

According to the modified Amontons' law,[59] the total lateral force, F_{LT} , could be described by:

$$F_{LT} = S_c \pi a^2 + \mu F_N \quad (2)$$

where S_c is the shear strength, a is the surface contact radius, μ is the friction coefficient. The first term is adhesion-controlled; its magnitude is governed by both the surface roughness and interactions between surfaces. The second term is normal force-controlled; its magnitude is determined predominantly by surface morphology and roughness.[52]

For a rigid surface, the surface contact radius, a , is small and does not increase significantly with the normal force, thus lateral force is mainly controlled by the normal force and increases linearly,[59] as seen for pure ILs in the boundary layer regime.[24] However, as shown in Fig. 6 (a), for OCP and -1.0 V the lateral force does not increase with the normal force linearly up to high force. This means the lateral force is significantly influenced by adhesion, [52,60–62] likely because the polymer chains in the adsorbed polymeric cation enriched boundary layers at such potentials are more compressed with the increase of normal load, leading to an increase of surface contact area.

The shear strength, S_c , and the surface contact radius, a , in the adhesion-controlled term could be estimated using different contact mechanics models. The Hertz model only considers hard wall repulsion at contact without taking into account of attractive forces at all.[63] The Derjaguin–Müller–Toporov (DMT) model

Table 1

The values of L_{contour} and L_K used for fits of the retraction force data shown in Fig. 5 to the modified freely-jointed chain model.^{54–55} Standard deviations are typically lower than $\pm 20\%$ for both L_{contour} and L_K .

	OCP					-1.0 V	+1.0 V							
L_{contour} (nm)	12	32	36	40	49	30	12	21	34	51	76	125	154	
L_K (nm)	0.10	0.12	0.12	0.12	0.13	0.12	0.13	0.13	0.13	0.13	0.13	0.13	0.13	

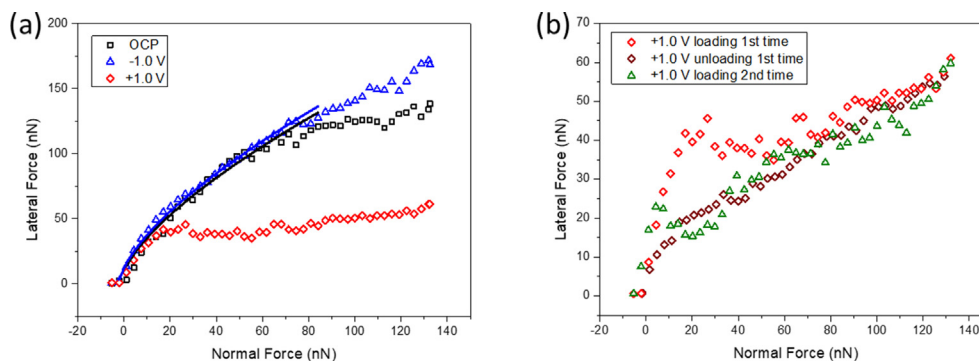


Fig. 6. (a) lateral force vs normal force of 5 wt% PIL/BMIM TFSI on Au(111) at OCP, -1.0 V and $+1.0$ V, respectively; (b) lateral force vs normal force of 5 wt% PIL/BMIM TFSI on Au(111) at $+1.0$ V when the normal force increases from 0 nN to 140 nN (loading), decreases from 140 nN to 0 nN (unloading) for the first time and increase again at the same position for the second time. Solid lines in (a) are fits to the JKR model.

Table 2

The adhesive force, F_A , obtained from AFM normal force measurements, $F_{A(\text{AFM})}$, and the JKR fits, $F_{A(\text{JKR})}$, the shear strength, S_c , and the surface contact radius, a , at a load of 70 nN from the JKR fits.

	$F_{A(\text{AFM})}$ (nN)	$F_{A(\text{JKR})}$ (nN)	$S_{c(\text{JKR})}$ (MPa)	$a(F_N = 70 \text{ nN}, \text{ nm})$
OCP	1.8 ± 0.9	1.8 ± 0.3	50 ± 5	62 ± 7
-1.0 V	2.6 ± 0.3	2.4 ± 0.3	19 ± 1	18 ± 1

works better for rigid materials that have lower surface energy and are resistant to deformation.[64] The Johnson–Kendall–Roberts (JKR) model suits soft materials with high surface energy and large deformation, such as polymers and hydrogels.[52,65] Therefore, the JKR model is used to estimate S_c and a in this work; the details are described in the SI.

Fig. 6(a) shows that JKR model could fit the lateral force data well up to a normal force of 70 nN for both OCP and -1.0 V, and the fitted adhesive force, F_A , is in good agreement with the measured values using AFM (Table 2).

Although the lateral forces for OCP and -1.0 V are similar up to a normal force of 70 nN, the JKR fitted S_c values are different (50 MPa and 62 MPa, respectively), as the adhesive forces used in the JKR fits are different, *c.f.* Table 2. However, the S_c values for both OCP and -1.0 V are obviously lower than the values (~ 135 MPa) observed previously on the bare gold surface.[66] As S_c is mainly determined by the interactions between the two contact surfaces including electrostatic interactions and H-bonding,[59] This agrees with our argument that the AFM probe slides on top of a polymeric cation-enriched boundary layer instead of the bare Au(111) surface at least up to a normal force of 70 nN. The fitted S_c is higher for -1.0 V than OCP. This indicates that polymeric cations are more enriched in the boundary layer at -1.0 V than at OCP, leading to stronger interactions with the slightly negatively charged silica AFM probe via electrostatic attraction and H-bonding of N–H with Si–O, similar to previous studies.[52] The surface contact radii, a , at a normal force of 70 nN are comparable at OCP and -1.0 V (Table 2). This indicates the chains in the polymeric cation-enriched boundary layers are compressed to similar degrees at both potentials, consistent with the normal force data in the previous section.

When the normal force is higher than 70 nN, the lateral force vs normal force data do not follow the JKR model any more for both OCP and -1.0 V, as shown in Fig. 6 (a). This suggests the cation-enriched boundary layers are fully compressed and the surface contact radii do not increase significantly; the lateral forces are mainly load-controlled. In this high normal force regime ($F_N > 70$ nN), the lateral force increases more slowly with normal force for OCP than for -1.0 V. Since the adsorbed boundary layer could be

brushed away at high normal force at OCP but not at -1.0 V (Fig. 3 (c) and (f), respectively), this suggests that BMIM cations could refill the boundary layer after the removal of the polymeric cations and lubricate the Au(111) surface at OCP.

It is clear that the addition of PIL to BMIM TFSI does not improve the lubricity at Au(111), especially at OCP and -1.0 V. However, the difference in lateral forces as a function of potential increases from ~ 15 nN ($F_N = 130$ nN) for pure BMIM TFSI (Fig. 2 (g)) and other conventional ILs[23–24,26] to ~ 125 nN for 5 wt% PIL/BMIM TFSI (Fig. 6 (a)). This offers an opportunity to apply the 5 wt% PIL/BMIM TFSI system as a smart “switch” to control advanced systems via actively tune the friction *in situ*, like next generation smart actuators, etc.

4. Conclusions

PILs have shown great potentials as flexible thin film electrolytes due to their good conductivity and mechanical characteristics.[27–29] The interfacial properties of PILs at electrode interfaces, including interfacial nanostructure and nanofriction, have great importance for applications in lubrication and electrochemistry; however, they have not been well known. In this work, we have found that when mixing 5 wt% PIL with one conventional IL BMIM TFSI, the polymeric cations can adsorb from the bulk liquid to the Au(111) electrode interface and form a boundary layer. This polymeric cation-enriched boundary layer binds strongly to the negatively charged surface, less strongly to the neutral surface and very weakly to the positively charged surface. The chains of the polymeric cations are compressed at -1.0 V and OCP due to electrical attractions with the electrode surface, whereas stretched at $+1.0$ V due to electrical repulsions. The lateral forces of the 5 wt % PIL/IL mixture at -1.0 V and OCP are higher than at $+1.0$ V and pure BMIM TFSI of the same potentials, owing to increased surface roughness and attractive interactions of the polymeric cation-enriched boundary layer with the AFM probe. At $+1.0$ V, the lateral force is much lower and comparable to pure BMIM TFSI due to the displacement of weakly adsorbed polymeric cation-enriched boundary layer with a more lubricating TFSI anion enriched boundary layer. Compared to previous investigated pure conven-

tional ILs[23–24,26], the difference in lateral forces as a function of potential increases by one order of magnitude for 5 wt% PIL/BMIM TFSI, which makes this system possible to work as a smart “switch” to control advanced systems via actively tune the friction *in situ*.

The results from this work significantly expand our understanding of PIL interfacial properties near Au(111) electrode surfaces at different electrical potentials and help to optimise surface-based applications of PILs, such as electrically tunable lubricants and flexible thin film electrolytes. The adsorption kinetics of the polymeric cation-enriched boundary layer as a function of potential will be probed in the future so as to elucidate the adsorption mechanisms of PILs better at the interface.

CRediT authorship contribution statement

Hua Li: Conceptualization, Methodology, Investigation, Formal analysis, Writing - original draft, Writing - review & editing, Visualization, Project administration, Funding acquisition. **Yunxiao Zhang:** Conceptualization, Methodology, Investigation. **Seamus Jones:** Conceptualization, Methodology, Investigation. **Rachel Segalman:** Conceptualization, Resources, Supervision, Writing - review & editing, Project administration, Funding acquisition. **Gregory G. Warr:** Conceptualization, Resources, Writing - review & editing, Project administration, Funding acquisition. **Rob Atkin:** Conceptualization, Resources, Supervision, Writing - review & editing, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jcis.2021.08.067>.

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